University of California, Berkeley

Print Form

Reset Form

EECS Department

GRADUATE STUDENT GENERAL PETITION FORM

TO BE COMPLETED BY STUDENT (PLEASE USE INTERACTIVE FORM)

FROM:	Almeda	Shm	Garanganao	SID#	3036239478
	LAST NAME	FIRST NAME	MIDDLE		

2838 Derby Street	Berkeley	CA	94705	shm.almeda@berkeley.edu
STREET ADDRESS	CITY	STATE	ZIP	EMAIL ADDRESS

STATE REQUEST & REASON FOR REQUEST BELOW:

Petition for CS294-176 (Readings in Neural Network Mathematics) taken Spring 2021 for AI Prelim Breadth Requirement
Syllabus and Course Schedule Attached.

STUDENT SIGNATURE — DATE: 4/26/2021

TO BE COMPLETED BY RESEARCH OR TEMPORARY ADVISOR

COMMENTS: I am supportive of this petition.
The instructor (Jonathan Shewchuck) should confirm if this course should be on the prelim list. Shm is making very good progress in her research.

RESEARCH/TEMP ADVISOR SIGNATURE — DATE: 4/28/21

TO BE COMPLETED BY DEPARTMENT

DECISION (CIRCLE ONE): **APPROVED** / ~~**NOT APPROVED**~~

CONDITION:

CHAIR OF GRADUATE MATTERS/PRELIM COMMITTEE SIGNATURE — DATE: 4/29/21

STUDENT NOTIFIED? (CIRCLE ONE) YES / NO

DATE

Rev. 6/30/05

CS294-176 Syllabus

GRADING

You'll be graded based on three things: you'll give one lecture presenting one or more papers/results; on Piazza, we'll discuss the papers and propose extensions and open problems; you'll choose one problem to try to make progress on and work on it. We'll allocate 100 points as follows:

40 points: Your presentation

- 35 points for the talk
- 5 points for the discussion and Executive Summary on the Piazza post you create for your presentation

30 points: Discussion (primarily on Piazza, but Zoom discussion might also earn credit)

- 20–30 points for proposing open problems or extensions of presented ideas (only 5 of these points can be for your own presentation)
- Up to 10 points for other forms of discussion of work presented by others.

30 points: Explore an open problem or an extension of a paper's result, and submit an essay about your ideas.

Presentations

- During the FIRST week of the semester, please choose a date for your presentation.
- You must schedule a dry run of your talk with me to take place at least 5 days before your class presentation. Expect to make revisions.
- At least 7 days before your class presentation, create a Piazza post stating the title and authors of the paper(s) you will discuss, and which particular sections or proofs you plan to discuss. Some ML papers are very long, in which case it's important to carefully specify to your classmates which parts you most want them to read—no more than you can discuss well in a 50 minute talk.
- At least 2 days before your class presentation, add a brief "Executive Summary" to your Piazza post stating the main take-home points you want your classmates to get out of the paper and your talk.
- Presentations should be 30–50 minutes long. Leave time for questions.

- Depending on paper length, I usually recommend not trying to present all the results in a paper. Instead, focus on conveying a few key mathematical intuitions so that everyone can take something away from your lecture.
- Presentations may be made with slides (e.g., Keynote, Powerpoint, PDF) or with pads and blackboard/whiteboard software, whichever best fits your style. You are welcome to borrow material from the original authors if those are available.
- You are responsible for answering your classmates' questions about your paper/presentation on Piazza.

Discussion

- There will be opportunities for oral questions during/after each presentation, but we will continue the discussion on Piazza. Most or all of your credit for discussion (30 points) will come from your Piazza contributions.
- A good habit for a researcher, when reading research papers, is to ask yourself how the paper might be extended, and what open questions are left unanswered. This will be a primary focus of our discussions (at least for the purpose of grading). After each presentation, try to think of extensions and add them to the corresponding Piazza post. (I don't expect you to comment on every presentation, but you should find something to say about at least half of them.)
- Any form of discussion (e.g., questions to the presenter) can earn up to 10 points, but the last 20 points (out of 30) are reserved for proposed extensions and open problems. (Being the first in the class to spot a major error in a paper also counts as an "extension" or "open problem", even moreso if you can fix it.)

Essay

- Choose one or two open problems or extensions to focus on. Try to develop some new ideas or solutions. Write an essay about it.
- Your essay should be at least one page long (not including title, author, references). Group collaborations are permitted, in which case your essay's length should be at least one page per author. There is no maximum length.
- If you work on several problems, you can submit several essays (summing to at least one page of ideas).
- I would be very happy if a few publications come out of this class, but it isn't required to get a good grade. The minimal standard is simply to explore an open problem or an idea for extending a paper's results: e.g., begin deriving the appropriate mathematics, develop some particular examples and explore their behaviors, make some conjectures, and everything else a researcher does to attack a problem.

LECTURES (Zoom Meeting ID)

Lectures will be on (irregular) Mondays and Wednesdays, 5:10–6:30 PM.

Meeting ID: 936 2214 2777
Passcode: 421551
https://berkeley.zoom.us/j/93622142777?pwd=bXB2TUxRQkgxSzk4WUttZUt5S2lIQT09

SIGN UP TO SPEAK!

I have created a Google spreadsheet where you can sign up to speak. Please choose a date no later than **Wednesday, January 27:**

https://docs.google.com/spreadsheets/d/1tymcfVs2KNw-UGxA-NHEJzbzrPpgKaSSaFcAbmecdu0/edit

You can choose a date before you decide what paper to talk about. First come, first serve.

At least **seven days** before your speaking date, please create a Piazza post giving the full title, author list, and link for the paper(s) you will present, so we can start reading. Please also tell us which particular sections or proofs you plan to discuss. Some ML papers are very long, in which case it's important to carefully specify to your classmates which parts you most want them to read—no more than you can discuss well in a 50 minute talk. Lastly, fill in the title and link in the spreadsheet above (so they're all in one place).

At least **five days** before your speaking date, please do a dry run of your talk with me. (In some circumstances less than five days might be okay, but leave adequate time for improvements!) Contact me by email to make arrangements.

At least **two days** before your class presentation, add a brief "Executive Summary" to your Piazza post stating the main take-home points you want your classmates to get out of the paper and your talk.

Talks should be 30-50 minutes long. If your paper is so cool you want to talk about it for more than 60 minutes, let me know and we can arrange a second speaking date.

READINGS

Here are some papers that might be worth covering in class (though I don't guarantee it!). I will try to add more to this list in the next few days. I have not ensured that the arXiv publication is the most recent version for all of these papers; I recommend checking to see if there's a newer version elsewhere before posting your paper choice.

Axiomatic Attribution for Deep Networks

Mukund Sundararajan, Ankur Taly, Qiqi Yan

https://arxiv.org/abs/1703.01365

Bounding and Counting Linear Regions of Deep Neural Networks

Thiago Serra, Christian Tjandraatmadja, Srikumar Ramalingam

https://arxiv.org/abs/1711.02114

Note: if you discuss this paper, you might also consider discussing Cover's Theorem; there's a good brief proof at https://www.cns.nyu.edu/~eorhan/notes/covers-theorem.pdf . Serra's paper cites a result by Zaslavsky, but I think it's the same as Cover's Theorem. There are two versions of Cover's Theorem, one for hyperplanes through the origin and one for general hyperplanes, and the corresponding results differ by a 1 somewhere, so look out.

The Multilinear Structure of ReLU Networks

Thomas Laurent, James von Brecht

https://arxiv.org/abs/1712.10132

On Transversality of Bent Hyperplane Arrangements and the Topological Expressiveness of ReLU Neural Networks

J. Elisenda Grigsby, Kathryn Lindsey

https://arxiv.org/abs/2008.09052

On the Optimization of Deep Networks: Implicit Acceleration by Overparameterization

Sanjeev Arora, Nadav Cohen, Elad Hazan

https://arxiv.org/abs/1802.06509

Exploring Generalization in Deep Learning

Behnam Neyshabur, Srinadh Bhojanapalli, David McAllester, Nathan Srebro

https://arxiv.org/abs/1706.08947

Theoretically Principled Trade-off between Robustness and Accuracy

Hongyang Zhang, Yaodong Yu, Jiantao Jiao, Eric P. Xing, Laurent El Ghaoui, Michael I. Jordan

https://arxiv.org/abs/1901.08573

The Marginal Value of Adaptive Gradient Methods in Machine Learning

Ashia C. Wilson, Rebecca Roelofs, Mitchell Stern, Nathan Srebro, Benjamin Recht

https://arxiv.org/abs/1705.08292

Certified Adversarial Robustness via Randomized Smoothing

Jeremy M Cohen, Elan Rosenfeld, J. Zico Kolter

https://arxiv.org/abs/1902.02918

VC Classes are Adversarially Robustly Learnable, but Only Improperly

Omar Montasser, Steve Hanneke, Nathan Srebro

https://arxiv.org/abs/1902.04217

Adversarially Robust Generalization Requires More Data

Ludwig Schmidt, Shibani Santurkar, Dimitris Tsipras, Kunal Talwar, Aleksander Mądry

https://arxiv.org/abs/1804.11285

Adversarial examples from computational constraints

Sébastien Bubeck, Eric Price, Ilya Razenshteyn

https://arxiv.org/abs/1805.10204

How Does Batch Normalization Help Optimization?

Shibani Santurkar, Dimitris Tsipras, Andrew Ilyas, Aleksander Madry

https://arxiv.org/abs/1805.11604

Robustness May Be at Odds with Accuracy

Dimitris Tsipras, Shibani Santurkar, Logan Engstrom, Alexander Turner, Aleksander Madry

https://arxiv.org/abs/1805.12152

Is Q-learning Provably Efficient?

Chi Jin, Zeyuan Allen-Zhu, Sebastien Bubeck, Michael I. Jordan

https://arxiv.org/abs/1807.03765

A Convergence Theory for Deep Learning via Over-Parameterization

Zeyuan Allen-Zhu, Yuanzhi Li, Zhao Song

https://arxiv.org/abs/1811.03962

Provably Robust Deep Learning via Adversarially Trained Smoothed Classifiers

Hadi Salman, Greg Yang, Jerry Li, Pengchuan Zhang, Huan Zhang, Ilya Razenshteyn, Sebastien Bubeck

https://arxiv.org/abs/1906.04584

Generalization and Equilibrium in Generative Adversarial Nets (GANs)

Sanjeev Arora, Rong Ge, Yingyu Liang, Tengyu Ma, Yi Zhang

https://arxiv.org/abs/1703.00573

Do GANs actually learn the distribution? An empirical study

Sanjeev Arora, Yi Zhang

https://arxiv.org/abs/1706.08224

Provable Bounds for Learning Some Deep Representations

Sanjeev Arora, Aditya Bhaskara, Rong Ge, Tengyu Ma

https://arxiv.org/abs/1310.6343

Neural Networks Should Be Wide Enough to Learn Disconnected Decision Regions

Quynh Nguyen, Mahesh Chandra Mukkamala, Matthias Hein

https://arxiv.org/abs/1803.00094

Neural Tangent Kernel: Convergence and Generalization in Neural Networks

Arthur Jacot, Franck Gabriel, Clément Hongler

https://arxiv.org/abs/1806.07572

On Exact Computation with an Infinitely Wide Neural Net

Sanjeev Arora, Simon S. Du, Wei Hu, Zhiyuan Li, Ruslan Salakhutdinov, Ruosong Wang

https://arxiv.org/abs/1904.11955

An Exponential Learning Rate Schedule for Deep Learning

Zhiyuan Li, Sanjeev Arora

https://arxiv.org/abs/1910.07454

Disentangling Feature and Lazy Training in Deep Neural Networks

Mario Geiger, Stefano Spigler, Arthur Jacot, Matthieu Wyart

https://arxiv.org/abs/1906.08034

Harmonic Networks: Deep Translation and Rotation Equivariance

Daniel E. Worrall, Stephan J. Garbin, Daniyar Turmukhambetov, Gabriel J. Brostow

https://arxiv.org/abs/1612.04642

Some of the papers listed on the following web page might also be appropriate:

https://unsupervised.cs.princeton.edu/deeplearningtutorial.html

Date	Name	Paper Title (shortened if long)	Link					
February 1	Sagnik Bhattacharya	Identity Matters in Deep Learning	https://arxiv.org/abs/1611.04231					
February 3	Sajant Anand	Deep Neural Networks as Gaussian Processes	https://arxiv.org/abs/1711.00165					
February 8	Dimitris Papadimitriou	Net-Trim: Convex Pruning of Deep Neural Networks with Performance Guarantee	https://arxiv.org/abs/1611.05162					
February 10	Alicia Tsai	Deep Equilibrium Models	https://arxiv.org/abs/1909.01377					
February 15	**Presidents' Day**							
February 17								
February 22	Rikhav Shah	Adversarial examples with small hamming distance	https://arxiv.org/abs/1901.10861					
February 24								
March 1	Chawin Sitawarin	Provable Robustness of ReLU networks via Maximization of Linear Regions	https://arxiv.org/abs/1810.07481					
March 3								
March 8								
March 10	Anders Christensen	On the Optimization of Deep Networks: Implicit Acceleration by Overparameterization	https://arxiv.org/pdf/1802.06509.pdf					
March 15								
March 17	Tony Zhao	Provably Robust Deep Learning via Adversarially Trained Smoothed Classifiers	https://arxiv.org/abs/1906.04584	* also covered Certified Adversarial Robustness via Randomized Smoothing				
	Spring Recess							
March 29								
March 31								
April 5	Rachel Lawrence	Adversarial examples from computational constraints	https://arxiv.org/abs/1805.10204					
April 7	Sid Kaushik	Axiomatic Attribution for Deep Networks	https://arxiv.org/pdf/1703.01365.pdf					
April 12	Andrew Ligeralde	Neural Networks Should Be Wide Enough to Learn Disconnected Decision Regions	https://arxiv.org/pdf/1803.00094.pdf					
April 14	Fanwei Kong	Theoretically Principled Trade-off between Robustness and Accuracy	https://arxiv.org/pdf/1901.08573.pdf					
April 19	Jamie Simon	Exponential expressivity in deep neural networks through transient chaos	https://arxiv.org/pdf/1606.05340.pdf					
April 21	Shm Garanganao Almeda	Do GANs actually learn the distribution? An empirical study	https://arxiv.org/abs/1706.08224					
(Please don't move later than April 21.)								